

For **robotics inverse kinematics (IK)**, many beginners think they need advanced university mathematics immediately. They don't.

Since your goal is **JARVIS → Walking Robot → ROS 2 → Quadruped → Manipulator Arm**, I would learn the topics in this order.

Level 1: Mandatory Mathematics (Learn Before IK)

1. Angles and Radians

You already have this in your roadmap.

Learn:

- Degrees $\leftrightarrow$ Radians
- Unit Circle basics

Questions:

- Convert 90° to radians.
- Convert $\pi/3$ radians to degrees.

2. Trigonometry

Learn:

- $\sin$
- $\cos$
- $\tan$
- inverse trig:
 - $\sin^{-1}$
 - $\cos^{-1}$
 - $\tan^{-1}$

These are used everywhere in IK.

Questions:

- If hypotenuse = 10 cm and angle = 30° , find the opposite side.

- Find θ if $\cos \theta = 0.5$.

3. Coordinate Systems

Learn:

- X-axis
- Y-axis
- Z-axis
- Local Frame
- Global Frame

Questions:

- Robot foot at (4,3). Distance from origin?
- Convert local coordinates to world coordinates.

4. Distance Formula

Questions:

- Find the distance between (2,3) and (5,7)?
- Distance from robot body to target point?

5. Vectors

Learn:

- Magnitude
- Direction
- Normalization

Questions:

- Magnitude of (3,4)?

- Normalize (6,8)?

Level 2: Mathematics Required for Real IK

6. Law of Cosines

This is the MOST IMPORTANT formula for beginner IK.

Suppose:

- Upper leg = 10 cm
- Lower leg = 10 cm

Target point:

- $x = 10$
- $y = 10$

Law of Cosines gives knee angle.

Questions:

- Find angle in triangle with sides 5,5,6.
- Why is IK impossible if target is beyond leg reach?

7. Law of Sines

Used occasionally for alternative IK derivations.

Questions:

- Find missing angle using Law of Sines.
- When is Law of Sines preferred over Cosines?

8. Matrix Basics

Not needed for v0.3.

Needed for:

- ROS
- TF Frames
- Robotics

Learn:

- Matrix multiplication
- Rotation matrix

Questions:

- Multiply 2×2 matrix by vector.
- Why use matrices for rotation?

Level 3: Robotics Mathematics

9. Forward Kinematics (FK)

Given:

- Joint angles

Find:

- Foot position

Questions:

- Servo = 30° , Knee = 60° . Where is foot?
- Why is FK easier than IK?

10. Inverse Kinematics (IK)

Given:

- Target position

Find:

- Joint angles

Questions:

- Foot must reach (10,10). Find angles.
- Why can one target have multiple IK solutions?

11. Homogeneous Transformation Matrices

Needed later for:

- ROS 2
- Manipulators
- Navigation

Not needed now.

Physics Needed for IK

The physics requirement is much smaller than the math requirement.

1. Torque

Learn:

- Lever arm
- Torque
- Moment arm

Questions:

- Why does a longer leg require more torque?
- Why is lifting weight horizontally harder?

2. Center of Mass

Learn:

- CoM
- Stability

Questions:

- Why do robots fall?
- How does battery placement affect stability?

3. Force

Learn:

- Newton's Laws
- Weight

Questions:

- Why can't servo lift unlimited loads?

- What happens when load doubles?

4. Inertia

Learn:

- Momentum
- Sudden stopping effects

Questions:

- Why does chassis shake?
- How does smoothing motion reduce shaking?

What YOU Actually Need Right Now

For your current roadmap:

Before v0.3

Learn:

1. Angles
2. Trigonometry
3. Coordinate Systems
4. Distance Formula
5. Vectors
6. Torque
7. Center of Mass

Before v0.6 (Real IK)

Add:

8. Law of Cosines

9. Law of Sines

10. Forward Kinematics

11. Inverse Kinematics

Before ROS Manipulators

Add:

12. Matrix Algebra

13. Transformation Matrices

14. Jacobians

15. Differential Kinematics